# Fourier For Ye

## Series, Transforms and More!

Ganesh

October 26, 2023

1 Throwback to Taylor Series

2 Fourier Series

3 Fourier Transforms

4 Applications

5 Discrete Signals

# Throwback to Taylor Series

What does the Taylor expansion of a function mean?

# Throwback to Taylor Series

What does the Taylor expansion of a function mean?

We approximate a function locally in the polynomial basis($\{x^n | n \in \mathbb{N}\}$).

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

since it is easier to work with polynomials.

https://imgur.com/gallery/dlTFGBb

# The polynomial basis

Just like $\{x^n | n \in \mathbb{N}\}$, it turns out that $\{1, \sin(2\pi nx), \cos(2\pi nx) | n \in \mathbb{N} - \{0\}\}$ also form a basis(an orthogonal one too!)

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But the function space must be periodic!

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# The Fourier decomposition

**Definition (Fourier series)**

The Fourier series of a $2\pi$ periodic function $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx)\right)$$

where,
$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) cos(kx) dx$
$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) sin(kx) dx$

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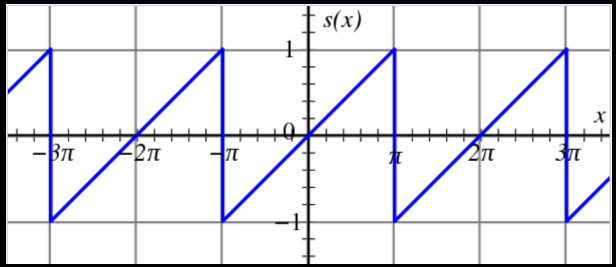
where,
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$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) sin(kx) dx$

$x$ is oftentimes the "time-domain" and $n$ the frequency. Notice that the frequencies here are integers(or integer multiples of 1; you will see why this is useful later).

# Example

$$f(x) = \frac{x}{\pi}, x \in [-\pi, \pi]$$

and $f(x + 2\pi) = f(x)$




Find the Fourier series?

# Example

Calculate $a_k s$ and $b_k s$

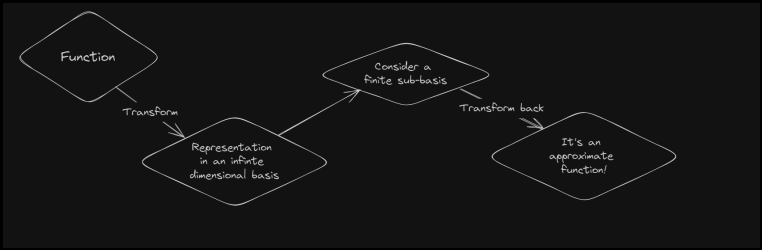
$$a_k = 0 \text{ and } b_k = \frac{2(-1)^{k+1}}{k\pi}$$

$$f(x) = \sum_{n=0}^{\infty} \left( \frac{2(-1)^{n+1}}{n\pi} \right) \sin(nx)$$

https://imgur.com/gallery/PIqiZQb

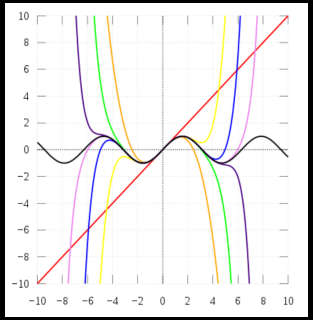
# Some caveats

- Just like the Taylor expansion, the infinite series represents the exact function, while a finite sum would be an approximation

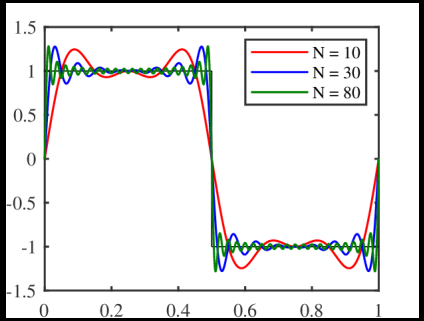


# Some caveats

- A better approximation means including more terms.




Taylor approximation




Fourier approximation

Do you notice any difference?

# Some caveats

- Unlike the Taylor series, Fourier series can handle discontinuous functions too!

Question: Why?

# Some caveats

- Unlike the Taylor series, Fourier series can handle discontinuous functions too!

Question: Why?

Loosely speaking, it's because Fourier decomposition depends on the integral of the function while the Taylor decomposition depends on the derivatives.

# Some caveats

- Fourier series for a function on $[-L, L]$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right)$$

where,
$a_k = \frac{1}{L} \int_{-L}^{L} f(x) \cos\left(\frac{k\pi x}{L}\right) dx$
$b_k = \frac{1}{L} \int_{-L}^{L} f(x) \sin\left(\frac{k\pi x}{L}\right) dx$

Notice that the "frequencies" are integer multiples of $\frac{\pi}{L}$

## Some caveats

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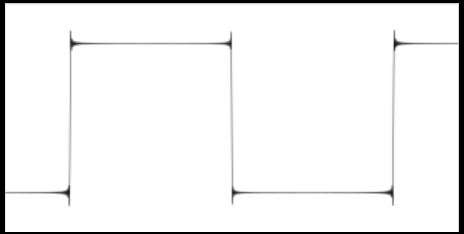
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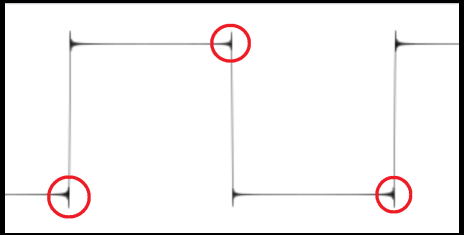
Similarly, one can extend the definition to the general domain $[a, b]$, $a, b \in \mathbb{R}$

# Gibbs phenomenon



This is the approximation of a square wave using 125 harmonics
Do you notice something off?

# Gibbs phenomenon



The function overshoots and undershoots around the points of discontinuities, and this "error" exists as long as you sum only a finite number of terms. In other words, it only converges "fully" when you consider the complete infinite sum.

Some keywords to explore: Pointwise, Uniform Convergence

# Complex Fourier series

The Fourier series can be re-written more concisely as

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{\pi i x n}{L}}, \quad i = \sqrt{-1}$$

where,

$$c_k = \frac{1}{2L} \int_{-L}^{L} f(x) e^{-i\frac{\pi k x}{L}} dx$$

# Complex Fourier series

The correspondence:

$$
\begin{aligned}
c_0 &= A_0 \\
c_k &= (A_k - iB_k)/2 \qquad \text{for } k > 0 \\
c_k &= (A_{-k} + iB_{-k})/2 \quad \text{for } k < 0
\end{aligned}
$$

The real and complex Fourier series are equivalent.
Note: These exponentials also form an identical orthogonal basis.

# Extending the definition

Can we somehow extend the definition of Fourier series to non-periodic function?

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# A neat trick

A non-periodic function is just...

# A neat trick

A non-periodic function is just...
a periodic function with $\infty$ period!

# Extending the definition

Consider the complex Fourier series of the $P$ periodic function $f(x)$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{\pi i x n}{L}}$$

where,

$$c_k = \frac{1}{2L} \int_{-L}^{L} f(x) e^{-i\frac{\pi k x}{L}} dx$$

Let $L \to \infty$

# Extending the definition

$$f(x) = \lim_{L\to\infty} \sum_{n=-\infty}^{\infty} \left(\frac{1}{2L}\int_{-L}^{L} f(x)e^{-i\frac{\pi nx}{L}}dx\right)e^{\frac{\pi ixn}{L}}$$

Let $y = \frac{n\pi}{L}$ and $\Delta y = \frac{\pi}{L}$. Thus we have the Riemann sum

$$f(x) = \lim_{L\to\infty} \sum_{n=-\infty}^{\infty} e^{ixy} \underbrace{\int_{-L}^{L} f(x)e^{-ixy}dx}_{\hat{f}(y)} \cdot \frac{\Delta y}{2\pi}$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(y)e^{ixy}dy$$

# Extending the definition

**Definition (Fourier transform)**

The Fourier transform of the function $f(x)$ is given by

$$\hat{f}(y) = \int_{-\infty}^{\infty} f(x) e^{-ixy} dx$$

Oftentimes $y$ is replaced with $\omega$ denoting frequency and $x$ with $t$ denoting time and is referred to as a map from the time domain to the frequency domain.

# Extending the definition

**Definition (The inverse Fourier transform)**

The inverse Fourier transform of the function $f(x)$ is simply

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(y) e^{ixy} dy$$

# Intuition

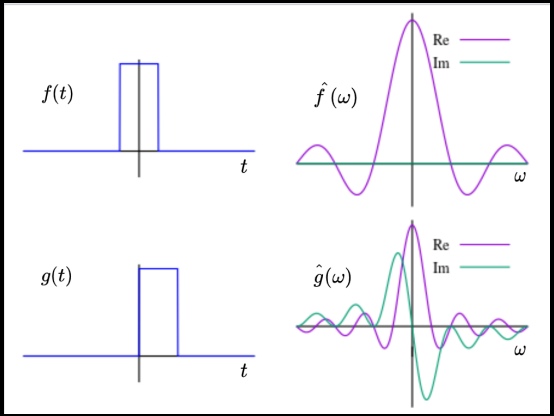
How does one interpret these formulas?

$$c_n \to \hat{f}(y)$$

$$\sum_{-\infty}^{\infty} \to \int_{-\infty}^{\infty}$$

We have transformed from a discrete set of frequencies to continuous ones.

# Examples




Notice the continuum of the frequency domain.

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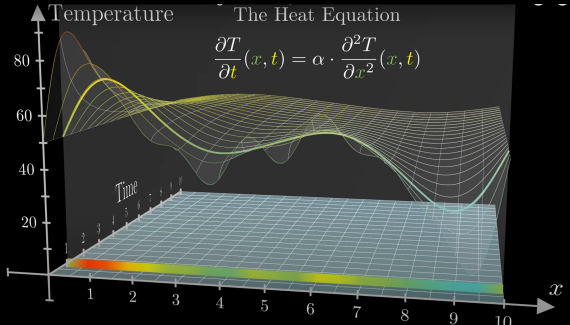
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# The Heat equation

## The actual problem that Fourier solved



# The Heat equation

The sine function solves the heat equation. Due to the linearity of the differential operator, a linear combination of sines would also be a solution.

So Fourier wanted to find a way to represent any heat distribution as a linear combination of sine waves, including common discontinuous heat distributions.

# Digression

The 2-D Fourier transform and its inverse of a function are given by

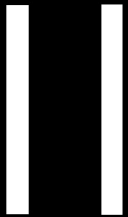
$$F(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)e^{-j2\pi(ux+vy)}dxdy$$

$$f(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(u,v)e^{j2\pi(ux+vy)}dudv$$

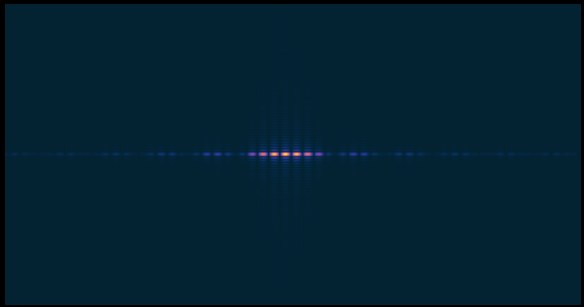
The Real valued 2D Fourier Series(not for the faint of heart)

# Diffraction

Let us perform a Fourier transform of an image



(a) Image



(b) Fourier transform

# Diffraction

The Fourier transform looks somewhat like the diffraction pattern.

# Diffraction

The Fourier transform looks somewhat like the diffraction pattern.
In fact the Fourier transform is **exactly** the diffraction pattern!

MATLAB simulation

# Diffraction

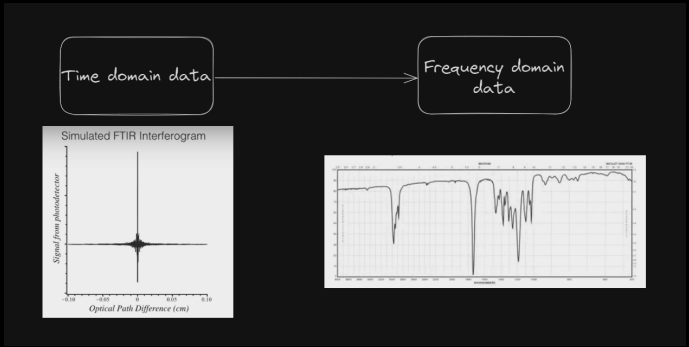
Proof: Consider an aperture in the $x-y$ plane to be a light source(with usual parameters $\lambda, k$) of the form $E_0(x_0, y_0)$. The field $E_i(x_i, y_i)$ at the screen sufficiently far away at a distance $z$ is given by the Fresnel integral:

$$E_i(x_i, y_i) = \frac{\exp\{(jkz)\}}{j\lambda z} \iint_{\text{Aperture plane}} E_0 \exp\left\{\frac{jk}{2z}\left[(x_i - x_0)^2 + (y_i - y_0)^2\right]\right\} dx_0\, dy_0$$

Simplify with appropriate assumptions to show that it is exactly the Fourier transform of the aperture illumination(Fraunhofer Diffraction).

# Spectroscopy

Fourier Transform spectroscopy: We study the characteristics of time domain spectrum through a frequency domain spectrum.




Keywords: Michelson interferometer, FTIR

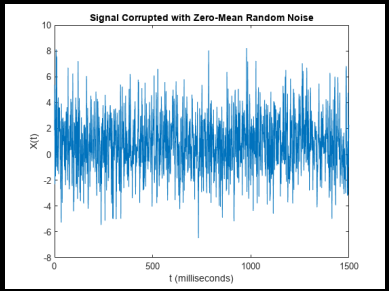
# Signal Processing

Fourier transform is extremely useful in processing signals

# Signal Processing

## De-noising

Consider a sound signal with some random noise added to it:




How does one remove this (high-frequency)noise? MATLAB page

# Signal Processing

## Image Processing

Filters:

- Edges: sharp changes. High frequency components.
- Smoothening: removing sharp changes/high frequency components.
- Low pass filter: Removes high-frequency components.
- High pass filter: Removes low-frequency components.

# Signal Processing

## Image processing



1 Throwback to Taylor Series

2 Fourier Series

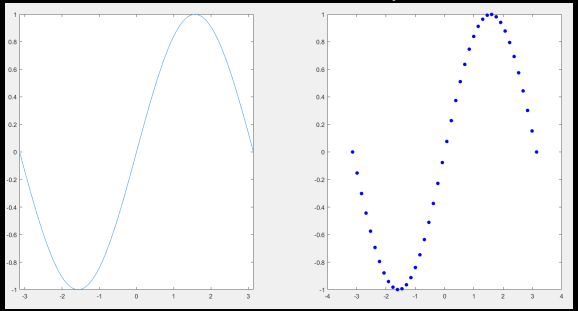
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# Discrete Fourier series

In real world, signals/functions are seldom continuous. We should therefore learn to deal with discrete data points.



# Discrete Fourier series

**Definition (Discrete Fourier Series)**

Given a list of $N$ equally spaced points $x(n)$, the Fourier series and its coefficients are given by

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{ik\frac{2\pi}{N}n}$$

and

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-ik\frac{2\pi}{N}n}$$

Proof: Involves proving *Nth* roots of unity form an orthogonal basis. Or define the discrete points using the Dirac delta as a piecewise continuous function and proceed the usual way.

# Discrete Fourier Transform

We are usually concerned with the Discrete Fourier Transform, which is often a misnomer for an operation that essentially is DFS. The term transform is misleading as there is no "integral" involved for it to be a "transform".

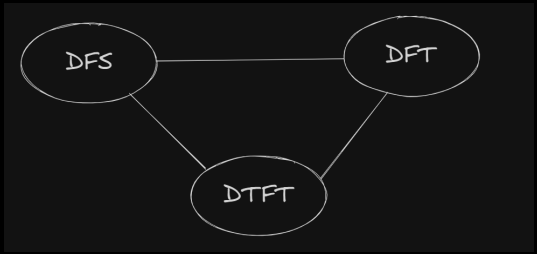
Discrete Fourier transform(DFT):

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-ik\frac{2\pi}{N}n}$$

Inverse DFT(IDFT):

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{ik\frac{2\pi}{N}n}$$

# The Various DFs



# DTFT

| | DTFT | DFT |
|---|---|---|
| input | discrete, infinite | discrete, finite |
| output | contin., periodic | discrete, finite |

# Sampling a continuous signal

How do we model a continuous signal with a discrete set of samples?

# Sampling a continuous signal

How do we model a continuous signal with a discrete set of samples?
How many samples sufficiently represent a signal?

# Nyquist-Shannon Theorem

**Theorem**

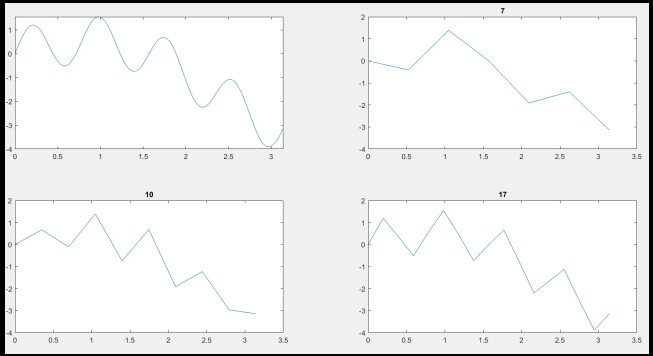
*If a function $x(t)$ contains no frequencies higher than $B$ hertz, then it can be completely determined from its ordinates at a sequence of points spaced less than $1/(2B)$ seconds apart.*

*(or)*

*If a function $x(t)$ contains no frequencies higher than $B$ hertz, then a sufficient sample rate is anything larger than $2B$ samples per second.*

# Nyquist-Shannon Theorem

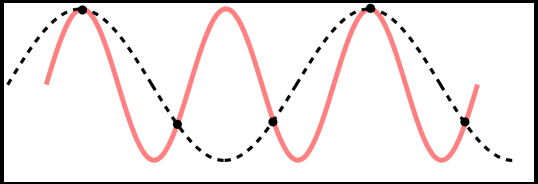
Consider the signal $S = \sin(8x) + x\cos(x)$ in $[-\pi, \pi]$




Different sampling rates

# Aliasing

Notice how under-sampling leads to another lower frequency wave:



Fixes: Adequate sampling, Low pass filtering.